

5.3 Exploring Rates

Lesson Introduction

 Benchmark	MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.		 Learning Targets	- I can use ratios and rates to describe relationships between quantities. - I can write a ratio as a unit rate.
 Benchmark Coherence	Building On		Working Toward	
	N/A		MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.	
 Essential Questions	- How do you use ratios and rates to describe relationships between quantities? - How do you write a ratio as a unit rate?			
 Lesson Narrative	In this lesson, students deepen their understanding of ratios. Students begin by representing ratios in different ways and connecting them to real-world contexts, and then connecting this understanding to rates and then unit rates. These new concepts are then applied to more real-world scenarios, first with units of measurements such as miles, and then extended to more abstract scenarios. The lesson culminates in a self-assessment to identify areas of struggle.			
 Component Summary	5.3.1 Warm-Up (~5 min.)	Students decide and justify which bag of chips they would rather share with a friend (<i>SE p. 184</i>)	5.3.4 Guided Practice (5 min.)	Practice using ratios and rates (<i>SE p. 186</i>)
	5.3.2 Collaboration (20 min.)	Work with a partner to express ratios as a to b , $a:b$, and $\frac{a}{b}$ (<i>SE p. 184</i>)	5.3.5 Your Turn! (5 min.)	Write unit rates from real-world scenarios (<i>SE p. 187</i>)
	5.3.3 Guided Instruction (10 min.)	Express ratios as rates and unit rates (<i>SE p. 185</i>)	5.3.6 Wrap-Up (~5 min.)	Red, Yellow, Green Light self-reflection (<i>SE p. 187</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Rate - Unit rate <u>Additional Terms:</u> N/A		(Optional) Note cards (Optional) Chart paper (Optional) Markers	- Create an anchor chart. - Consider bringing in some measuring cups and/or bags of chips and containers of sauce to help students visualize the size of the quantities they are working with in the problems.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may forget to simplify ratios with division before multiplying to find the unknown unit. - Students may invert ratios when trying to find an equivalent rate. - Students may struggle to understand differences between fractions and ratios and what they represent.	- Consider creating an anchor chart and allowing students to add to their reference cards with instructions and examples to aid them in using ratios and rates to describe relationships between quantities and writing a ratio as a unit rate. - Remind students to label the units in the quantities and then to keep the unit placement the same in the second ratio. This will help avoid mistakes.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 5.3.1 Warm-Up provides an engaging entry point for students to debate with a friend. Use the “Would You Rather?” prompt to position students as claim-makers, using evidence with at least one number in their responses to prove who might have the better answer. There is no one right answer, so encourage creativity in responses.	MA.K12.MTR.2.1: Multiple Representations Throughout the lesson students see ratios represented multiple ways. The ratios are expressed three ways: a to b , $a:b$, and $\frac{a}{b}$. The ratios are represented using tables, double number lines, and tape diagrams. Students draw connections among these representations throughout the lesson.
 Differentiate Instruction	The Wrap-Up asks students to write which topics they feel stuck on, are starting to get the hang of, and feel confident in. Consider allowing students to choose whether to record themselves explaining the answer verbally, drawing and labeling a diagram, or providing an example.	
Lesson Notes:		

5.3.1 Warm-Up: Would You Rather?

Component Overview: Students compare and justify which bag of chips they would rather share with a specified number of friends.

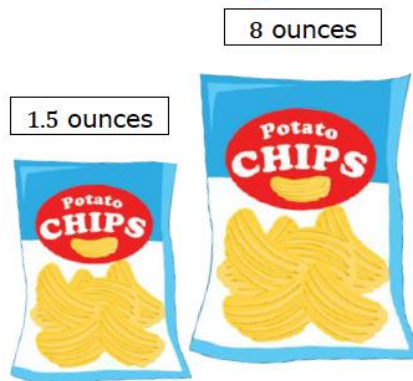


5.3.1 Warm-Up: Would You Rather?

1. Would you rather share the small bag of chips with 1 friend or the large bag of chips with 8 friends? Explain your reasoning using mathematics.

Answers will vary.

I would rather share the bigger bag since it would give more chips per person than the smaller bag.



Think-Pair-Share

Ensure that students realize two people (one friend and you) are sharing the chips in the first scenario, and nine are sharing the chips in the second scenario. Consider positioning students to discuss question prompts before or instead of writing their answers. Use sentence stems to encourage conversation such as, “I would rather share the ... bag with one friend because...” and “I would rather share the ... bag with eight friends because...”

5.3.2 Collaboration: Expressing Ratios in Different Ways

Component Overview: Students work with a partner or small group to express ratios as a to b , $a:b$, and $\frac{a}{b}$. Once students have completed the two questions, share answers as a class and discuss how each group arrived at their response.

5.3.2 Collaboration: Expressing Ratios in Different Ways

Ratios are often expressed in the real world as an amount of "something per something." For example, in setting the world record, Joey Chestnut ate 7.4 hot dogs per minute.

Discuss each of the following scenarios with your partner and work together to answer the questions.

1. A recipe for marinara sauce requires 56 ounces (oz) of whole peeled tomatoes. The recipe yields 8 servings of marinara sauce.
 - a. What is the ratio of tomatoes to sauce?
56 ounces of tomatoes to 8 servings of sauce.
56 to 8, 56:8 or $\frac{56}{8}$
 - b. Complete the tape diagram to represent the ratio of tomatoes to sauce.

tomatoes (ounces)	7	7	7	7	7	7	7	7
sauce (servings)	1	1	1	1	1	1	1	1

- c. How many ounces of whole peeled tomatoes are in 1 serving of marinara sauce? *7 oz.*
 - d. Complete the ratio.

$$\frac{7 \text{ oz of tomatoes}}{1 \text{ serving}}$$



Students may discuss whether each box should have 7 ounces or 8 servings. For students who are unsure, consider prompting questions.

- Why are there 8 boxes? What do they represent?
- How would you determine the equal amount in each of the 8 servings for a total of 56 ounces?
- What if there were 14 ounces in 2 servings? How many ounces would be in each serving?

5.3.2 Collaboration: Expressing Ratios in Different Ways (continued)

2. In 2012, there were 982 farms in Hardee County, Florida. The total amount of farmland was 273,978 acres.

- a. Write the ratio of acres : farms.

273,978:982, or $\frac{273,978 \text{ acres}}{982 \text{ farms}}$, or 273,978 to 982

- b. Complete the statement.

The total amount of farmland in Hardee County is split up among all the farms, so to determine the average number of acres for 1 farm,

- add 982 to 273,978.
- subtract 982 from 273,978.
- multiply 273,978 by 982.
- divide 273,978 by 982.

- c. Complete the ratio.

$$\frac{279 \text{ acres}}{1 \text{ farm}}$$

- d. Discuss with your partner whether you think this number of acres per 1 farm means that every farm in Hardee County was that size in 2012. Then, write a brief summary of your reasoning.

Answers will vary. No, because not everyone will purchase the same amount of land. Some people may be able to afford 1 acre whereas others may be able to afford 279 acres.



Students should be concluding that they must divide to determine the ratios for each of these questions. Listen for students who are making the connection that because ratios are multiplicative, this means they can divide to determine the ratio per 1. Highlight this thinking before moving onto the next component, as it will prime students for the discussion of unit rates.

5.3.3 Guided Instruction: Understanding Rates

Component Overview: This Guided Instruction introduces students to rates and unit rates.

5.3.3 Guided Instruction: Understanding Rates

Each of the ratios discussed so far in this lesson are also **rates**.

 A **rate** is a ratio that compares two quantities of different units.

A ratio is a comparison of two numbers, so all rates can also be expressed using ratios. Rates are often used to express how long it takes to do something.

1. Consider a familiar scenario involving a rate.
Tyler ran 2 miles in 30 minutes.
- a. Draw a tape diagram for the ratio that represents Tyler's run.

Miles	1 mile	1 mile
Minutes	15 minutes	15 minutes

- b. Complete the ratios. Then, complete each statement to interpret the ratios.

$$\frac{\frac{1}{15} \text{ miles}}{1 \text{ minute}}$$

$$\frac{15 \text{ minutes}}{1 \text{ mile}}$$

Tyler runs $\frac{1}{15}$ miles for every 1 minute.

Tyler runs 15 minutes for every 1 mile.

- c. Discuss with your partner how to use one of the rates to determine how many miles Tyler runs in an hour. Then, complete the statement.
Tyler runs 4 miles per hour.

Each of the rates used to describe Tyler's run is a **unit rate**.

 A **unit rate** is a ratio comparing a number of units of one quantity to 1 unit of a second quantity.



Turn and Talk

After the definitions are read out loud, have students participate in a Turn and Talk routine to synthesize each definition. Students should not be reading the definition or parroting what is already in the box, but instead using their own words to demonstrate understanding. If students are struggling to define in their own words, encourage them to come up with an example of a rate and unit rate.



Students may struggle to understand the difference between rate and unit rates or may not comprehend the significance of being able to calculate both. Encourage students to consider why knowing the unit rate may be more helpful in certain scenarios.

How is a unit rate different than a rate?

Why is it important to know the unit rate of comparison?

When might it be helpful to use the unit rate?

5.3.4 Guided Practice: Using Ratios and Rates

Component Overview: Students represent the relationship between various ingredients and calories on a nutrition label using ratios. Consider partnering students in pairs during this component.



5.3.4 Guided Practice: Using Ratios and Rates

1. A nutritional label for raspberry ice cream is shown.

Nutrition Facts		
3 servings per container		
Serving size		2/3 Cup (123g)
Calories	Per Serving	Per Container
		1000
		% Daily Value*
Total Fat	58g	74%
Saturated Fat	36g	180%
Trans Fat	1.5g	
Cholesterol	365mg	128%
Sodium	550mg	24%
Total Carbohydrate	106g	39%
Dietary Fiber	0g	0%
Total Sugars	68g	
Includes Added Sugars	72g	144%
Protein	16g	
Vitamin D	3mcg	15%
Calcium	400mg	39%
Iron	2mg	19%
Potassium	532mg	19%

- a. What is the ratio of calories to servings?
1,000: 3

- b. What is the ratio of total carbohydrates in the container to the number of calories in the container?
106: 1,000

- c. What is the ratio of added sugars to the number of servings?

72: 3

- d. Complete the unit rate to express the number of added sugars per serving.

24 grams
1 serving



Consider bringing in a variety of snacks or have one student per group bring in their group's favorite snack to share while working. Students can demonstrate their understanding using their favorite snack in substitution of or in addition to this label provided.



Challenge students who finish early to find the unit rates of each relationship.

5.3.5 Your Turn!

Component Overview: Students work independently to write unit rates from real-world scenarios.



5.3.5 Your Turn!

1. Tracy McGrady is a former NBA basketball player from Bartow, Florida. He played in 938 games in the NBA, scoring a total of 18,381 points. Complete the ratio to express the number of points he scored per game as a unit rate. Round the number of points to the nearest tenth.

$$\frac{19.6 \text{ points}}{1 \text{ game}}$$

2. The ruby-throated hummingbird is the most common hummingbird found in Florida. This bird can flap its wings 1,590 times in 30 seconds. Complete the ratio to express how fast the ruby-throated hummingbird flaps its wings per second.

$$53 \text{ wing flaps} : 1 \text{ second}$$



Remind students to label the units that correspond to each quantity in the ratio and then to keep the unit placement the same in the second ratio. This will help avoid mistakes.

What does “per” mean?

How did you find the points per game?

How did you find the wing flaps per second?

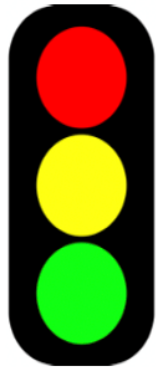
5.3.6 Wrap-Up: Reflection

Component Overview: Students use a stoplight reflection to consider which topics they feel stuck on, are starting to get the hang of, and feel confident with.



5.3.6 Wrap-Up: Reflection

1. Complete the stoplight reflection to share how you are feeling about the concepts learned in today's lesson.



I still feel stuck about how to ...

Answers will vary. If students are completely stuck they may choose this option.

I'm starting to get the hang of ...

Answers will vary. If students understand it but need more practice they may choose this option.

I feel confident and ready to go with ...

Answers will vary. If students are completely confident in this topic they may choose this option.



For students who might struggle verbalizing their understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can be used for students to reflect.